LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to main plane, aero dynamic and structural considerations, loads on wings, basic structural requirements, role of wings as stru
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to main plane, aero dynamic and structural considerations, loads on wings, basic structural requirements, role of wings as structure, positioning of components, form of wing construction
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of wing section model to identify spars, ribs, and skin; verification of load path understanding; instructor supervision required for model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the main plane?
(b) Name two types of loads acting on a wing.
(c) How does the wing contribute to overall aircraft stability?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Aerodynamic and structural considerations • Loads on wings: lift, drag, weight, thrust • Basic structural requirements for wings • Role of wings as primary structure • Positioning of components on the wing	L & Ds	PPT, IB	15 min
(b)	• Form of wing construction: Monospar vs. Multi-spar • Rib and Spar configuration details • Skin types and attachment methods • Fuel tank integration in wing structure • Control surface mounting points	L & Ds	PPT, PDF	20 min
(c)	• Structural failure modes in wings • Maintenance inspection points for wings • Correlation between load and material choice • Impact of wing positioning on aircraft balance • Summary of wing structural elements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the wing spars resist bending moments.
(b) Describe the relationship between wing loading and structural thickness.
(c) Identify three key structural elements visible on the wing model.

12. Home Assignment and Follow Up	Prepare a brief note comparing the structural layout of the main wing to the empennage structure covered in LP 13, focusing on load distribution differences.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Empennage structure, configuration of longitudinal stabilizer and concord and fore plane, layout of configuration of lateral stabilizer and control	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Instructor, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min